

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title: COMPLETED RESEARCH		Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 13

(1) Research Program/Project/Study Title: FIELD PERFORMANCE OF COCONUT SEEDLINGS GROWN FROM EXCISED EMBRYOS	
Author's Name:	Salvacion J. Legaspi
Academic Rank:	Professor VI
Co-Authors (with academic rank):	Raymund Lizada Heralyn Paderes Felomino Llorico
Campus/Satellite College/Address:	CAPSU - BURIAS CAMPUS
Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Pontevedra						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Dragon Fruit		(8) Discipline/Field of Specialization							

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 13

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture Project of Capiz State University Burias Campus to evaluate and document the field performance of the different varieties of coconut seedlings grown from excised embryos. The chosen excised embryos from 2 to 4 inches and 8 to 12 inches were transplanted in the field. Initial and current data was gathered and presented in the graph. Coconut seedlings from excised embryos measuring 8 to 12 inches displayed varying growth in terms of plant height, stem girth, total number of leaves, and present number of leaves produced. It was found that the field performance of the different varieties of excised coconut seedlings, when transplanted in the field, showed a comparable response regardless of the age of the trees. Among all the coconut varieties excised at 2 to 4 inches, only Kenabalan variety was found to perform well when transplanted in the field.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The palm tree is a tree of life, and while it yields many things, copra is still its principal byproduct. Numerous trees in the previous two years have been devastated by typhoons and, more recently, by the Brontispa infestation. Therefore, extensive tree restoration and replacement are required. Producing seed nuts in order to do this required giving up the ability to create copra.

In the Philippine setting, other significant issues facing the coconut industry include the degeneration of coconut trees, the widespread use of poor or low-yielding coconut varieties as a result of a lack of high-quality coconut seedlings, poor agronomic or farm management practices, the unrestricted removal of coconut trees due to strong marketability, poor soil nutrition, the occurrence of pests and diseases, natural disasters, the conversion of coconut lands, and a lack of sustained and adequate resources for infrastructure (L. Arhin et al., 2023; Gurbuz & Manaros, 2019).

The low pricing of coconut production and goods poses an overwhelming impediment. Coconuts are mostly farmed by smallholder farmers with less than four hectares of land. They are among the Philippines' poorest farmers. The extreme fluctuation of pricing makes it even more difficult to make a livelihood from coconut agriculture. Furthermore, the complicated nature of the coconut supply chain leads to significant logistical and transportation expenses in the Philippines, in which farmers have been tolerable for years (Strengthening Sustainable Coconut Production in the Philippines | Cargill, n.d.).

To address such hurdles, the Department of Agriculture has initiated strategic plans and programs through its Coconut Farmers and Industry Development Plan (CFIDP), which aims to boost the revenue and productivity of the 2.5 million Filipino coconut growers. It will also assist them and their families in alleviating poverty, obtaining an education, achieving social

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 13

fairness, and modernizing the whole coconut sector (Jampac, J.P., 2023).

On the one hand, studies on the physical attributes of excised coconut seedlings at various stages of growth and their survival rate have been previously studied and revealed that the diameter of the historium, volume of left-over liquid endosperm, number of roots and stem girth were affected by variety and storage of excision of the coconut seedling and from the germinating nut (Legaspi, 2019). This is being supported by the study of Molina M. (2010), which states that the initial length and circumference of roots and survival rate of dwarf coconut were found to be significant. Also, coconut seedlings can be excised from the germinating nuts and can be utilized as planting material. Overall findings revealed much more lateral roots and significantly longer shoots in seedlings whose roots had been removed (Thuzar et al., 2022).

Furthermore, an investigation into the optimum period of retention in a coconut variety of MAWA (Malayan Dwarf x West African Tall), for instance, revealed that coconut seedlings planted in polybag and field nurseries demonstrated that due to their size, the twelve and fourteen-month-old palms were more difficult to handle and experienced more transplanting shock than younger palms. Planting of ten-month-old palms grown in polybags is preferred with careful consideration for management aspects, the degree of transplanting shock, and early field performance (Krishnakumar et. al., 2019).

In a 2018 research by the Crop Science Research and Development Center, it was shown that coconut seedlings could be retrieved from the germination of coconut nuts with varying heights and their survival rates were only moderately high. However, further studies on the field performance of coconut should be ascertained.

It is within this premise that the coconuts' field performance at various stages of growth and their survival rate needs to be further investigated.

OBJECTIVES OF THE STUDY

The study was conducted to evaluate the field performance of coconut seedlings grown from excised embryos. It specifically aimed to:

1. Evaluate the field performance of coconut seedlings grown from excised embryos.
2. Document the field performance of coconut seedlings grown from excised embryos.

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted at the Organic Agriculture Project of Capiz State University, Burias Campus, Mambusao, Capiz. The initial data was gathered from 8 to 12

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 13

inches of excised coconut seedlings before transplanting in the field. Current data was also gathered from excised coconut seedlings that survived in the field. The parameters gathered were plant height, stem girth, total number of leaves, and present number of leaves. The total number of leaves were retrieved from the sum of the leaf scars and the present leaves were all manually counted. All the data gathered was presented using bar graph for clarity. The data were gathered in monthly basis.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant Height of Different Excised Coconut Seedlings

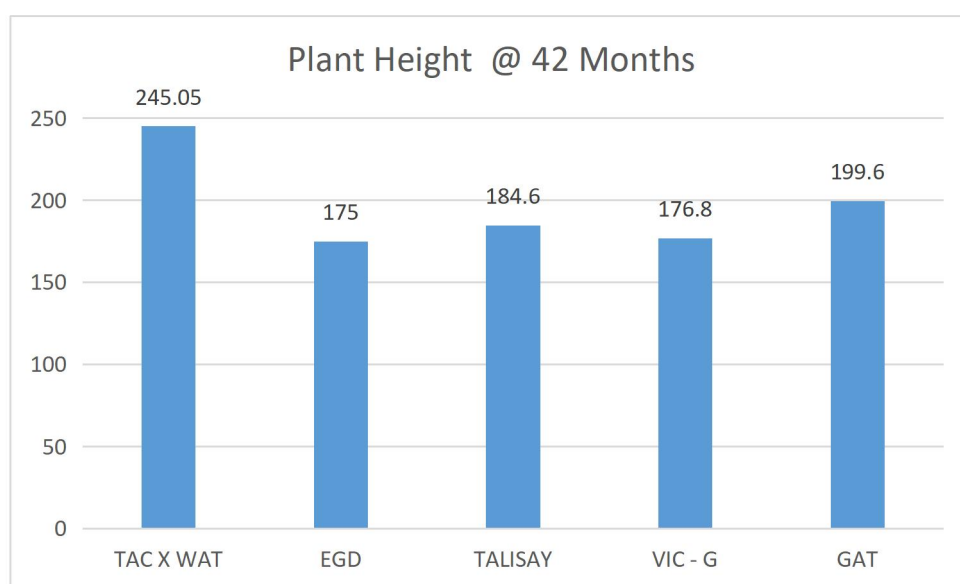


Figure 1. Plant height of different excised coconut seedlings forty-two months after transplanted in the field.

The table presents the different plant heights of the different excised coconut seedlings at 42 months, 39 months, and 28 months. Even though coconut seedlings had varied pre-transplant growth rates, they all displayed diverse plant development when transplanted into the field.

In forty-two (42) months or three years and six months, the coconut variety TAC x WAT produced the highest plant height (245.05 cm). This was followed by GAT (199.96 cm), Talisay (184.6), and VIC-G (176.8). While the EGD variety produced the shortest height (175 cm.).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 13

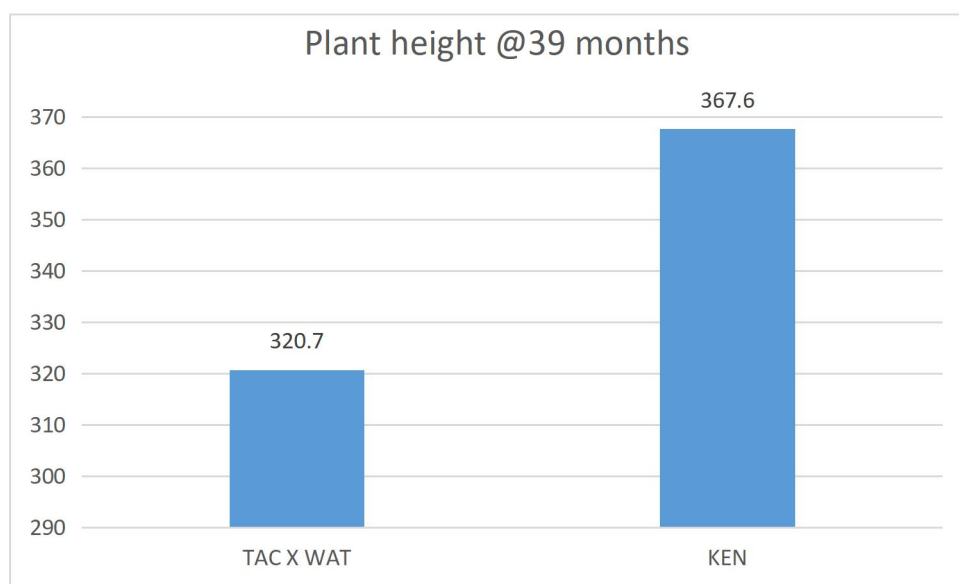


Figure 2. Plant height of different excised coconut seedlings thirty-nine months after transplanted in the field.

In thirty-nine (39) months, or three years and six months, the KEN variety produced the highest plant height measuring 367.6 centimeters. While the coconut hybrid variety TAC x WAT attained the shortest plant height (320.7).

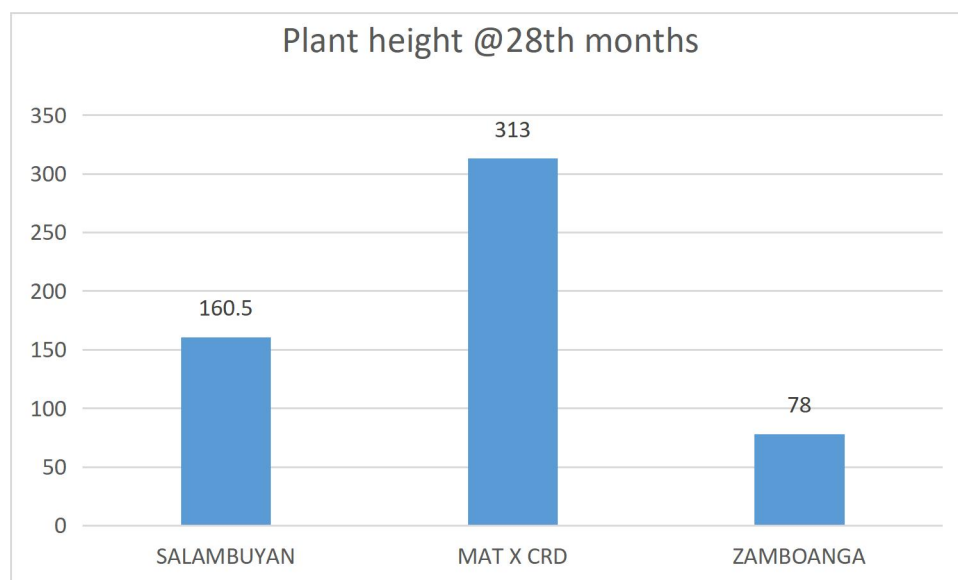


Figure 3. Plant height of different excised coconut seedlings twenty-eight months after transplanted in the field.

At twenty-eight months, the hybrid variety MAT x CRD demonstrated the highest plant

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 13

(313 cm.), which was followed by Salambuyan (160.5) and Zamboanga attained the shortest plant height (78 cm.) among the three.

Stem girth

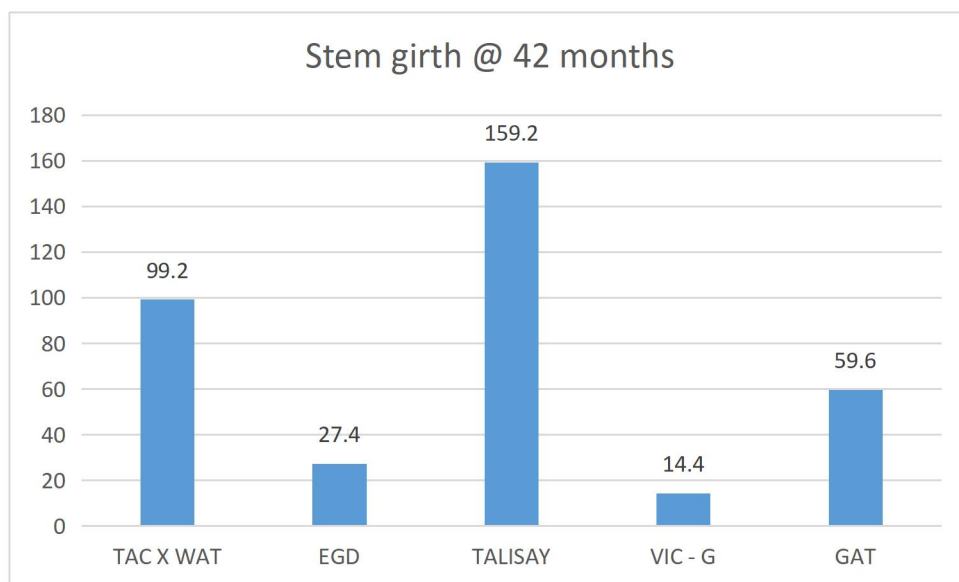


Figure 4. Stem girth of different excised coconut seedlings forty-two months after transplanted in the field.

The table demonstrates the stem girth of the different excised coconut seedlings planted in the field. Different coconut varieties of excised coconut seedlings showed varying attributes in their field performance as indicated.

In the forty-two months after field transplantation, the biggest stem girth was obtained by the Talisay variety (159.2 cm), which was followed by the hybrid coconut TAC x WAT (99.2 cm), EGD (27.4 cm.) and GAT (59.6 cm.). While VIC-G variety produced the smallest stem girth (14.4 cm).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 13

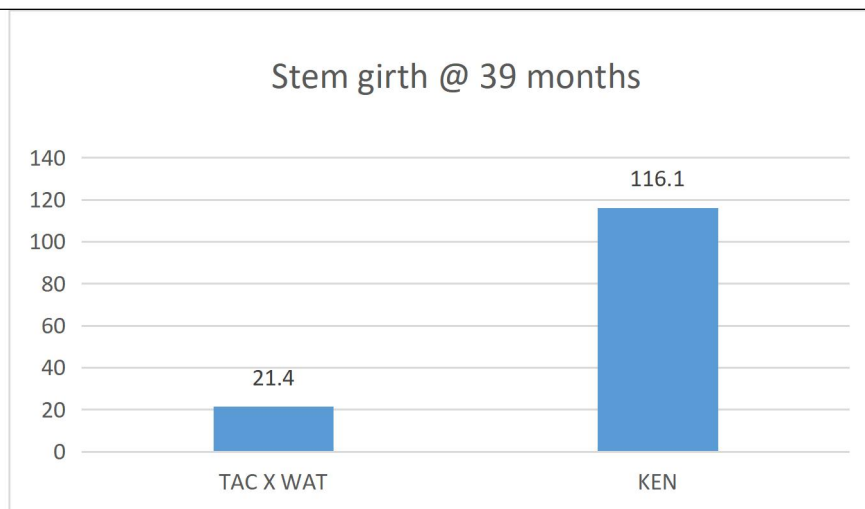


Figure 5. Stem girth of different excised coconut seedlings at thirty-nine months after transplant in the field.

The different excised coconut seedling varieties produced varying stem growth as exhibited in their stem girth. In terms of stem girth, thirty-nine months after being transplanted in the field, the KEN variety revealed an increased stem girth (116.1 cm), while the TAC x WAT had the shortest stem girth (21.4 cm.)

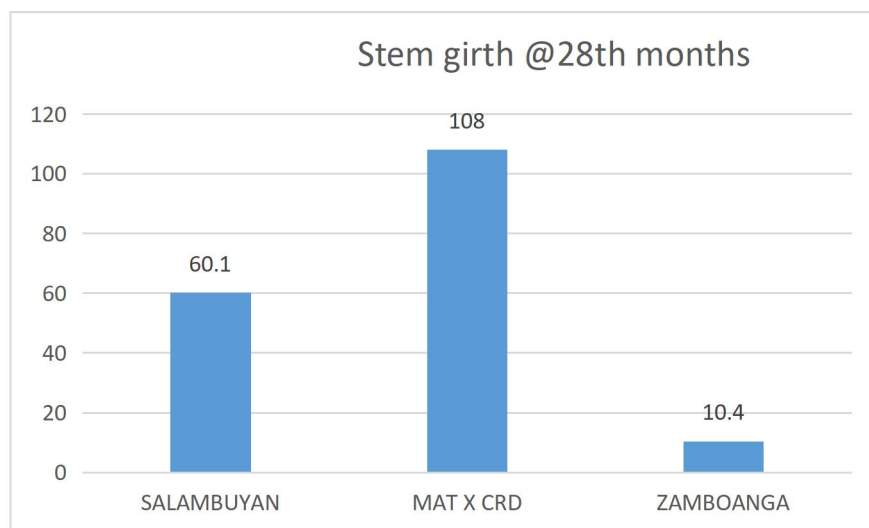


Figure 6. Stem girth of different excised coconut seedlings at twenty-eight months after transplant in the field.

Twenty-eight months after being transplanted in the field, the hybrid variety MAT x CRD obtained an increased stem girth (108 cm), followed by the Salambuyan variety (60.1 cm) while the Zamboanga variety had the shortest stem girth (10.4) among the three varieties.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 13

Total Number of Leaves

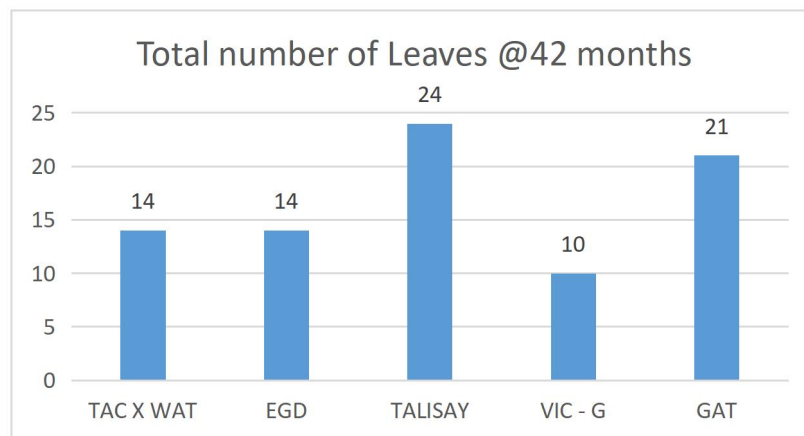


Figure 7. Total number of leaves produced by the different excised coconut seedlings at forty-two months after transplanted in the field.

The following results document the field performance, in terms of leaves produced, by the different excised coconut varieties specifically after forty-two, thirty-nine and twenty-eights months of field transplantation.

During three years and six months after transplanting in the field, the Talisay variety produced a total number of twenty-four (24) leaves, which was followed by the GAT variety with twenty-one (21) total number of leaves. While TAC x WAT and EGD varieties generated equal number of leaves (14). On the other hand, VIC-G attained the least number of total leaves (10).

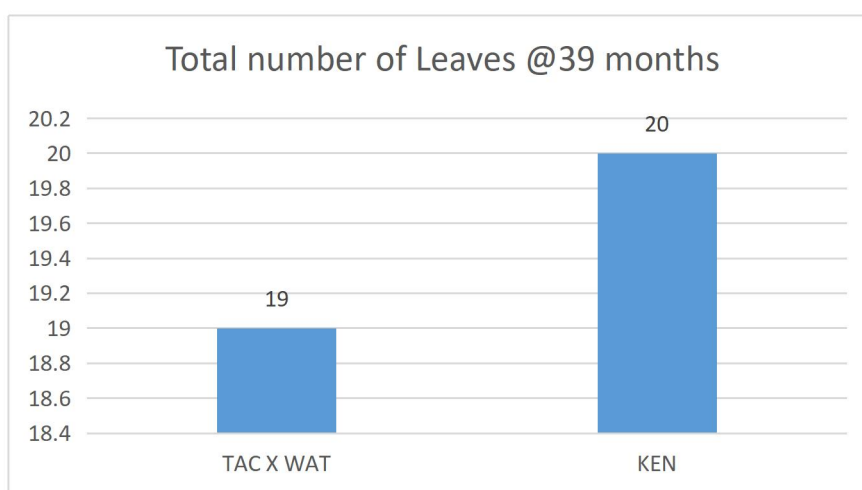


Figure 8. Total number of leaves produced by the different excised coconut seedlings at thirty-nine months after transplanted in the field.

The graph presents the total number of leaves produced after three years and three months after being transplanted in the field. Among the two varieties, the Kenabalan variety produced a total

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 9 of 13

number of twenty (20) leaves, while TAC x WAT had nine-teen (19) total leaves.

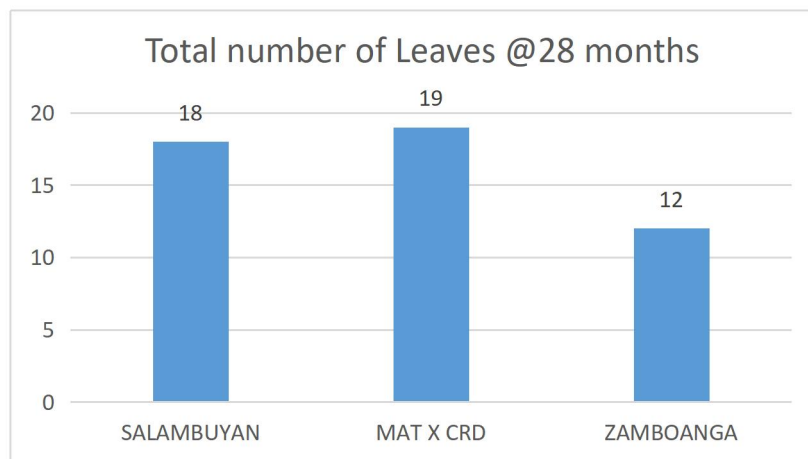


Figure 9. Total number of leaves produced by the different excised coconut seedlings at twenty-eight months after transplanted in the field.

Among the three varieties, during twenty-eight months after being transplanted in the field, MAT x CRD variety obtained nine-teen (19) total number of leaves, which was then followed by Salambuyan (18) and Zamboanga variety (12).

Present Number of Leaves

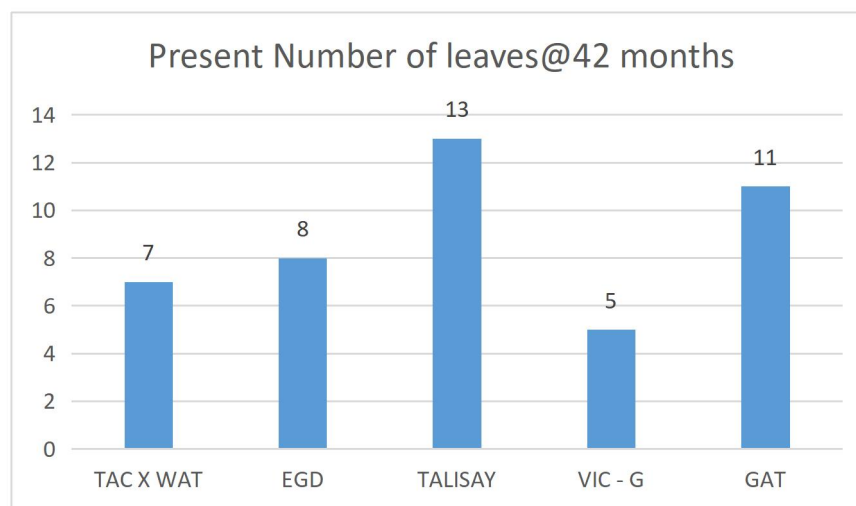


Figure 10. Present number of leaves of the different excised coconut varieties at forty-two months after transplanted in the field.

The following results document the field performance, in terms of present number of leaves produced, by the different excised coconut varieties specifically after forty-two, thirty-nine and twenty-eights months of field transplantation.

After three years and six months of transplanting in the field, thirteen (13) leaves were present from

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 10 of 13

the Talisay variety. GAT variety with eleven (11) leaves, EGD with eight (8) leaves, TAC x WAT with seven (7) leaves and VIC-G with five (5) leaves.

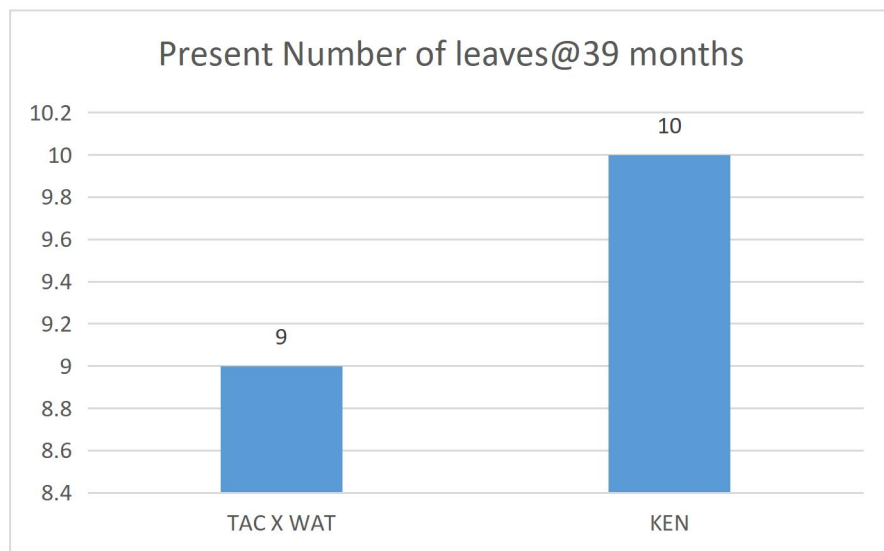


Figure 11. Present number of leaves of the different excised coconut varieties at thirty-nine months after transplanted in the field.

After three years and three months of transplanting in the field, the Kenabalan variety presently have ten (10) leaves while the hybrid variety TAC x WAT have nine (9) leaves.

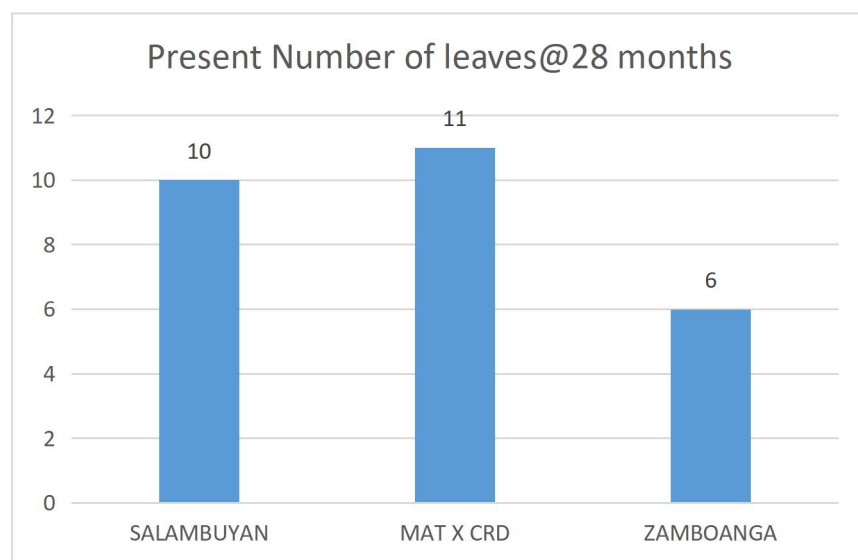


Figure 12. Present number of leaves of the different excised coconut varieties at twenty-eight months after transplanted in the field.

After two years and four months of transplanting in the field, among the three varieties, the MAT x CRD variety presently have eleven (11) leaves, followed by the Salambuyan variety with ten (10)

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 11 of 13

leaves, while the Zamboanga variety have six (6) leaves.

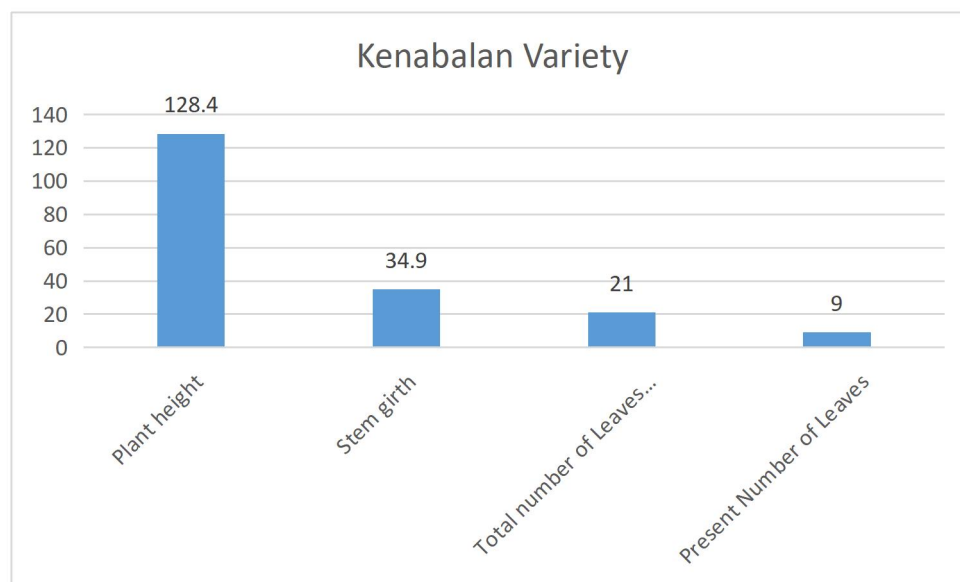


Figure 13. Growth performance of Kenabalan dwarf variety excised at 2- 4 inches embryo after forty-two months.

The table presented the growth performance of Kenabalan dwarf variety after being transplanted in the field. In terms of plant height, the Kenabalan recorded an increased height measuring 128.4 cm. An increase of 34.9 cm was noted in the stem girth, nine leaves were now present, and generated a total of twenty-one (21) leaves.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the result of the study, the following conclusions were obtained:

1. Excised coconut seedlings from embryos measuring 8 to 12 inches showed varying growth in terms of plant height, stem girth, number of leaves produced and number of present leaves. The coconut varieties displayed diverse plant development when transplanted in the field.
2. Hybrid coconut varieties are adaptive when transplanted in the field. Despite the differing age of transplantation in the field, coconut varieties TAC x WAT, KEN, and MAT x CRD show comparable performance as reflected in their plant height.
3. Stem girth of the excised coconut seedlings was influenced by months after

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 12 of 13

transplantation in the field.

4. The total number of leaves produced differs in the different varieties of excised coconut seedlings.

RECOMMENDATIONS:

The following recommendations are forwarded based in the results obtained:

1. Continue the observation until such time that a sufficient amount of seed nuts are available.
2. Follow the recommended cultural practices to prolong and improve the coconut seedlings' field performance.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 13 of 13

Arhin, L., Abdullah, S. N. A., Jaafar, J. N., & Ramlee, S. I. (2023). Conventional and modern breeding technologies for improving dwarf coconut cultivars: A review. *Journal of Horticultural Science & Biotechnology*, 1–12. <https://doi.org/10.1080/14620316.2023.2195863>

Gurbuz, I. B., & Manaros, M. M. (2019). IMPACT OF COCONUT PRODUCTION ON THE ENVIRONMENT AND THE PROBLEMS FACED BY COCONUT PRODUCERS IN LANA DEL. ResearchGate. Accessed through https://www.researchgate.net/publication/342158289_IMPACT_OF_COCONUT_PRODUCTION_ON_THE_ENVIRONMENT_AND_THE_PROBLEMS_FACED_BY_COCONUT_PRODUCERS_IN_LANA_DEL_NORTE_PROVINCE_PHILIPPINES

New program to provide more support to coco farmers | ATI Caraga. (n.d.). ATI Caraga. Accessed through <https://ati2.da.gov.ph/ati-13/content/article/jason-paul-dads-e-jampac/new-program-provide-more-support-coco-farmers>

Strengthening sustainable coconut production in the Philippines | Cargill. (n.d.). Accessed through <https://www.cargill.com/sustainability/strengthening-sustainable-coconut-production-in-the-philippines>

Thuzar, M., Sae-Lee, Y., Saensuk, C., Pitaloka, M. K., Dechkrong, P., Aesomnuk, W., Ruanjaichon, V., Wanchana, S., & Arikrit, S. (2022). Primary Root Excision Induces ERF071, Which Mediates the Development of Lateral Roots in Makapuno Coconut (*Cocos nucifera*). *Plants*, 12(1), 105. <https://doi.org/10.3390/plants12010105>.

Krishnakumar, V., Thampan, P. K., & Nair, M. A. (2019). *The Coconut Palm (Cocos nucifera L.) - Research and Development Perspectives*. Springer.

Molina Marie May (2010). Growth and Survival Rate of Excised Coconut Seedlings at Different Stages of Growth.

Moyin-Jesu, E. I. (2014). Comparative Evaluation of Different Organic Fertilizer Effects on Soil Fertility, Leaf Chemical Composition and Growth Performance of Coconut (*Cocos nucifera* L.) Seedlings. *International Journal of Plant and Soil Science*, 3(6), 737–750. Accessed through <https://doi.org/10.9734/ijpss/2014/4338>